Final Year Project: Deliverables Specification

Simulation & Scheduling of Parallelisable Tasks Across Local Devices and MEC Infrastructure

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Mandatory

- Literature review of task scheduling, offloading, parallelism in MEC, and use of local devices as part of a MEC system
- Review of previous FYPs in the same area (Cian Boyle, James Finn, Luke King)
- Simple implementation of a MEC simulator with three basic levels (cloud, edge, local device) alongside a central scheduler for parallelisable tasks
- Verification and validation that the simulator works properly (e.g. a 100 MI subtask takes 1 second on a 100 MIPS processor, and other similar tests)
- A basic first implementation of a scheduling algorithm in the simulator with fixed parameters and a small number of devices and homogeneous subtasks

Discretionary

- Improvement to the algorithm to consider more users
- Extension of the simulator to allow for more flexibility e.g. different processing speeds i.e. heterogeneous processors, multiple tasks per user, etc.
- Adaptation of the algorithm to work under dynamically changing conditions

Exceptional

- Enhanced algorithm that considers parallelisable tasks of a more complex nature, for example, heterogeneous subtasks or dependencies between subtasks